

Noah Bean

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Education

Oregon State University

B.S. Electrical and Computer Engineering

GPA: 3.81/4.0

Corvallis, OR

December 2025

Technical Skills

- **Languages & Low-Level:** C, Embedded C, C++, Python, Assembly (ARM, RISC-V, x86), SystemVerilog
- **Microcontrollers & RTOS:** STM32, ESP32, ARM Cortex-M, FreeRTOS, Zephyr, Bare-Metal, CMSIS, interrupt-driven drivers, power management, bootloaders, DSP, MISRA, Agile
- **Embedded Linux & BSPs:** Buildroot, U-Boot, Yocto/BitBake, Linux Kernel, device drivers, systemd
- **Protocols & Interfaces:** UART, SPI, I²C, CAN, Modbus, MQTT, BLE, TCP/IP
- **Hardware Validation & Debug:** Oscilloscopes, logic analyzers, JTAG/SWD, protocol analyzers, GPIB/VISA, TDR, unit/regression testing, test automation, Static Analysis
- **Tools & PCB Design:** STM32CubeIDE, ESP-IDF, Xilinx Vivado, KiCad, Altium, GCC, CMake, Git, TDD, OOP

Professional Experience

Langmuir Systems

Embedded Systems Engineer

Conroe, TX

July 2026 – Present

- Developed and debugged embedded C++ firmware for CNC/laser manufacturing equipment.
- Performed hands-on hardware debugging and failure analysis on PCB assemblies and embedded control systems.

Collins Aerospace

Systems and Electrical Engineering Intern

Wilsonville, OR

June 2025 – December 2025

- Performed functional and electrical verification of flight-critical avionics hardware, operating mixed-signal test benches with oscilloscopes, power supplies, and protocol analyzers to validate hardware behavior against system specifications.
- Developed Python test automation frameworks interfacing with bench instrumentation.

Intel Corporation

Electrical Engineering Intern

Hillsboro, OR

April 2024 – September 2024

- Performed root-cause analysis on signal integrity failures in server-grade CPUs, GPUs, and DDR5 memory subsystems using oscilloscopes and logic analyzers to isolate timing and power delivery issues in high-volume silicon validation.
- Built Python tooling to automate extraction and correlation of on-chip telemetry, fault logs, and signal integrity metrics with Machine Learning, cutting manual failure analysis overhead by 30%.

Projects

Air Quality Monitor

STM32 • FreeRTOS • Modbus RTU

C, CMSIS

- Architected a multi-task FreeRTOS firmware on STM32F446RE with five concurrent tasks covering sensor acquisition, system health monitoring, error handling, Modbus RTU communication, and data logging.
- Implemented a custom Modbus RTU slave protocol from scratch with interrupt-driven UART, CMSIS register-level drivers, and synchronization primitives (queues, semaphores, mutexes); integrated watchdog timer for fault-tolerant operation.

ESP32 IoT Weather Station

ESP32 • ESP-IDF • FreeRTOS • AWS IoT Core • KiCad

C

- Designed a custom PCB in KiCad for the ESP32-C3-02 SoC, integrating a TP4056 Li-ion charging circuit with under-voltage lockout, USB-C power input, BME280 environmental sensing, I2C OLED display, and SD card storage; produced full schematic, layout, BOM, and Gerber output.
- Developed multi-task FreeRTOS firmware with DHT22 sensing, dual-mode Wi-Fi (AP + Station), MQTT telemetry to AWS IoT Core over TLS, HTTP web server with OTA support, NVS credential storage, and SNTP time sync; distributed tasks across both cores with message queues, event groups, and binary semaphores for ISR signaling.

Custom Embedded Linux Image

Buildroot • U-Boot • Linux Kernel 6.18 • systemd

C, Shell

- Implemented complete BSP bring-up for TI AM335x (ARM Cortex-A8): configured U-Boot 2026.04, compiled Linux 6.18 kernel with Device Tree, generated root filesystem via Buildroot defconfig, and integrated systemd services.
- Developed and cross-compiled a custom Buildroot package containing a C binary interfacing with the kernel via `sysinfo()` syscall and `/proc` parsing; deployed as a systemd oneshot service with persistent journal logging.